

# Artur lasenovets

Information Security Research Engineer

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|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Profile                 | 4+ years of experience in software and security engineering. I develop in Go, Python, Rust, and TypeScript, with some experience in C++. I translate cryptographic protocols from papers into code, conduct security, correctness and leakage analysis, and develop systems for private data search, retrieval, and processing based on homomorphic encryption. |                                            |
| Education               | <b>Chongqing University of Posts and Telecommunications</b> , Chongqing, China                                                                                                                                                                                                                                                                                  |                                            |
|                         | ◇ <b>M.S.</b> in Network and Information Security                                                                                                                                                                                                                                                                                                               | Sep. 2024–Jul. 2027 ( <i>expected</i> )    |
|                         | ◇ Supervisor: <a href="#">Prof. Tang Fei</a>                                                                                                                                                                                                                                                                                                                    |                                            |
|                         | ◇ GPA: 3.98/4.00                                                                                                                                                                                                                                                                                                                                                |                                            |
|                         | <b>Ukhta State Technical University</b> , Ukhta, Russia                                                                                                                                                                                                                                                                                                         |                                            |
|                         | ◇ <b>B.S.</b> in Information Systems and Technologies                                                                                                                                                                                                                                                                                                           | Mar. 2022–Jun. 2024                        |
|                         | ◇ GPA: 4.49/5.00                                                                                                                                                                                                                                                                                                                                                |                                            |
|                         | <b>St. Petersburg State University of Telecommunications</b> , St. Petersburg, Russia                                                                                                                                                                                                                                                                           |                                            |
|                         | ◇ <b>B.S. studies</b> in Information Systems and Technologies                                                                                                                                                                                                                                                                                                   | Sep. 2020–Mar. 2022 ( <i>transferred</i> ) |
| Publications            | <b>Published Works and Preprints</b>                                                                                                                                                                                                                                                                                                                            |                                            |
|                         | <a href="#">Artur lasenovets</a> , Fei Tang, H. Zhu, P. Wang, and L. Liu. “ <a href="#">Bringing Private Reads to Hyperledger Fabric via Private Information Retrieval</a> ,” arXiv:2511.02656, 2025. Manuscript under review.                                                                                                                                  |                                            |
|                         | H. Zhu, F. Tang, W. Qin, J. Shan, P. Wang, and <a href="#">Artur lasenovets</a> . “ <a href="#">EVMK-SSE: Efficient and Verifiable Multi-Keyword Symmetric Searchable Encryption</a> ,” <i>Journal of King Saud University – Computer and Information Sciences</i> , 2025.                                                                                      |                                            |
|                         | <a href="#">Artur lasenovets</a> and Fei Tang. “ <a href="#">MFCTIIF: Multi-Feed Cyber Threat Intelligence Integration Framework</a> ,” <i>International Research Journal</i> , no. 1 (163), 2026.                                                                                                                                                              |                                            |
|                         | Piyumi M. S. Rajapaksha*, <a href="#">Artur lasenovets*</a> , Yinxue Yi, and Fei Tang. “TWAPPA-FRL: Trust-Weighted Privacy-Preserving Aggregation for Federated Reinforcement Learning in Collaborative Intrusion Detection,” 2026. Manuscript under review. *Equal contribution.                                                                               |                                            |
| Awards and Achievements | <b>Chinese Government Scholarship (CSC) for Graduate Studies</b>                                                                                                                                                                                                                                                                                                |                                            |
|                         | Full scholarship for the entire M.S. programme at CQUPT.                                                                                                                                                                                                                                                                                                        |                                            |
|                         | <b>Sber Student Accelerator Regional Demo Day</b>                                                                                                                                                                                                                                                                                                               |                                            |
|                         | Completed the Sber Student Accelerator with the AcademAI team and participated in the North-Western District regional demo day.                                                                                                                                                                                                                                 |                                            |
|                         | <b>Winner, EdTech Project Competition</b>                                                                                                                                                                                                                                                                                                                       |                                            |
|                         | Presented LLM-based educational platform and won the competition organized by the MIPT Phystech School, the Fund for Development of Phystech Schools, and Startech.Base.                                                                                                                                                                                        |                                            |
|                         | <b>MIPT &amp; Sber Phystech Gigachat Challenge</b>                                                                                                                                                                                                                                                                                                              |                                            |
|                         | Participated in the Phystech Gigachat Challenge Hackathon, developing a LLM-based educational platform using GigaChat and Kandinsky APIs within a 48-hour sprint.                                                                                                                                                                                               |                                            |
|                         | <b>Kaspersky Cyberimmunity Hackathon 2.0</b>                                                                                                                                                                                                                                                                                                                    |                                            |
|                         | Participated in Kaspersky’s Cyberimmunity Hackathon in Fall 2023, placing 11th out of 100+ teams for developing traffic monitoring solutions for low-altitude airspace security.                                                                                                                                                                                |                                            |
|                         | <b>First Degree Diploma, Severgeoecotech</b>                                                                                                                                                                                                                                                                                                                    |                                            |
|                         | Awarded first place in the Modern Information Technologies section of the 23rd International Youth Scientific Conference.                                                                                                                                                                                                                                       |                                            |
|                         | <a href="#">Supporting certificates</a>                                                                                                                                                                                                                                                                                                                         |                                            |

## Experience

### Graduate Researcher @ [CQUP](#)

Sep. 2024–Present

- ◇ Designed and implemented BGV-based private information retrieval in Go using Lattigo for private world-state queries in Hyperledger Fabric, with query privacy under an HBC peer model based on the IND-CPA security of BGV.
- ◇ Designed the TWAPPA-FRL architecture and implemented CKKS-based private aggregation of DQN model updates using TenSEAL, with IND-CPA-based client-update privacy under an HBC aggregator model and explicit leakage analysis.
- ◇ Analyzed EVMK-SSE's threat model with an HBC data owner and user and a malicious cloud server; demonstrated query privacy in the real/ideal model based on OT security and the pseudorandomness of CI-rOPRF/PRG under explicitly defined search- and access-pattern leakage.
- ◇ Implemented and validated an FPS/RFPS-style private pattern-search prototype, replacing the DFT-based matching transform under approximate-arithmetic CKKS with an NTT-based matching transform under exact-arithmetic BGV/BFV, with IND-CPA-based pattern privacy under an HBC server model and explicit protocol leakage.
- ◇ Reconstructed preprocessing models, query construction, server computation, and response expansion of PIR protocols such as Piano, FrodoPIR/SimplePIR, XPIR, SealPIR, and OnionPIR, and analyzed their communication/performance trade-offs.

### Information Security Engineer @ [Positive Technologies](#)

Aug. 2024–Feb. 2025

- ◇ Built an automated malware-processing pipeline with Airflow DAG that collects samples from MalwareBazaar, stores artifacts in S3, and scans them with YARA.
- ◇ Developed Python tools with unit-test coverage for validating Windows paths, network configurations, and endpoint-control policies.
- ◇ Analyzed EVTX logs, PowerShell, and malicious VBA macros; extracted IoCs and mapped TTPs to MITRE ATT&CK for CVE-2020-5902 and CVE-2023-38831.

### Software Engineer @ [AcademAI](#)

Sep. 2023–Aug. 2024

- ◇ Developed the application layer with Next.js/TypeScript application, Python/Flask services, PostgreSQL/Prisma data layer, and Auth.js authentication for an LLM-based educational platform used by computer science university faculty and students.
- ◇ Integrated LangChain-based generation workflows with AsyncIO and Redis for asynchronous LLM orchestration and intermediate state management.
- ◇ Containerized the application with Docker, configured Nginx, deployed to cloud-hosted VPS infrastructure, and built the GitHub Actions CI/CD workflow.

### Junior Software Engineer @ [ZVEK "Progress" LLC](#)

May. 2022–Aug. 2023

- ◇ Built a C++17/Crow REST backend with MySQL and Qt 5/QML desktop and mobile clients for a QR-based warehouse inventory system.
- ◇ Supported approximately 1,000 records and 20+ concurrent users; added unit-test coverage, CMake builds, Git-based development, debugging, and Linux deployment.

## Skills

- ◇ **Programming:** Go, Python, Rust, TypeScript, C++, SQL, Bash, LaTeX
- ◇ **Privacy-enhancing technologies:** Private Information Retrieval (PIR/CPiR), Searchable Symmetric Encryption (SSE), Private Pattern Search
- ◇ **Homomorphic encryption:** Brakerski–Gentry–Vaikuntanathan (BGV), Brakerski–Fan–Vercauteren (BFV), Cheon–Kim–Kim–Song (CKKS)
- ◇ **Post-quantum cryptography:** ML-KEM, ML-DSA, SLH-DSA; LWE, RLWE
- ◇ **Cryptographic libraries:** Lattigo, SEAL, TenSEAL, OpenFHE, fheanor
- ◇ **Systems and infrastructure:** Linux, Proxmox VE, VMware, Docker, GitHub Actions, GitLab CI/CD, Nginx, Ansible, Airflow, S3, Hyperledger Fabric
- ◇ **Security engineering:** YARA, Chainsaw, malware/log analysis, MITRE ATT&CK
- ◇ **Languages:** Russian — native; English — [C1](#); Chinese — [HSK3](#)